hydrodynamic and thermodynamic processes is also a rewarding task for validating numerical simulations in fundamental research and for guiding applications.

The complexity of the sequence of breakdown-induced events in different parameter regimes and under different boundary conditions imply a similar complexity of modeling tasks. First attempts have been made to use artificial intelligence for physical model-guided machine learning for accelerating laser induced plasma-mediated micro-machining process optimization [404]. Based on a modular system of different physical modeling tools and on data collected in systematic experiments and during material processing, artificial intelligence could help to select and combine appropriate modules for different regions of the parameter space.

6.3. Exploration of low-density plasma effects below the breakdown threshold

In the present paper, we have focused attention to nonlinear energy deposition by individual laser pulses with energies above the breakdown threshold. However, when series of tightly focused pulses with small energies are applied at high repetition rates, free electrons can produce material modifications also below the cavitation bubble threshold [22]. Here, in the low-density plasma regime, the laser action changes from mechanical to chemical effects. Figure 16 presents the irradiance range and the corresponding range of free-electron densities defining the low-density plasma regime. It covers seven orders of magnitude from the irradiance at which free electrons start to produce photodamage in nonlinear microscopy [405-409] up to the cavitation bubble threshold with single pulses.

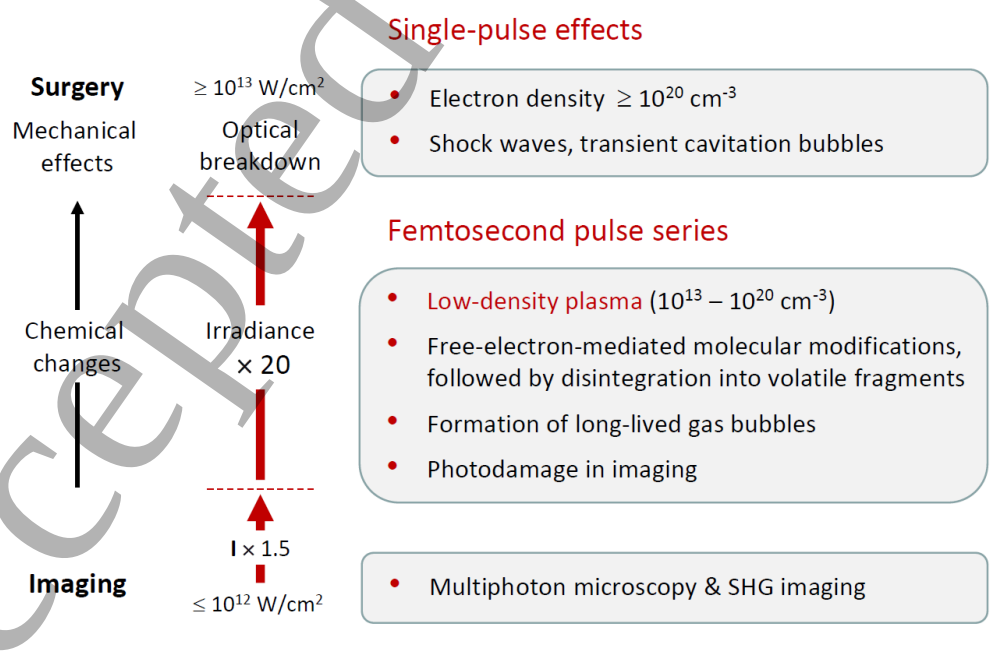

Figure 16 The realm of low-density plasmas between irradiance values employed for nonlinear imaging with fs laser pulse series and the optical breakdown threshold above which cavitation-associated laser surgery is performed.

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Nonlinear microscopy is performed at irradiance values between 10^{11} W/cm^2 and 10^{12} W/cm^2 [410]. Photodamage sets in already ≈ 1.5 times above the irradiance level that is regularly used for multiphoton imaging [407]. Here, the free-electron-yield becomes comparable to the nonlinear photon yield [406]. At about 20 times larger irradiance, a vapor bubble can be produced by a single laser pulse [37]. In the low-density plasma regime, the dominant laser-material interactions are free-electron-mediated chemical changes. In the transition regime just below the cavitation threshold, thermoelastic stresses still play a role for tissue and cell surgery, in conjunction with free-electron-mediated bond breaks [17,22]. However, at lower irradiance, the heating by individual pulses or pulse trains is far too small to cause any thermal changes [411]. Here, direct chemical bond breaking by free electrons goes hand in hand with indirect effects via dissociation of water molecules forming $^{\bullet}$OH radicals, which then interact with biomolecules [198]. Molecular fragmentation produces long-lived bubbles of noncondensable gas. Gas bubble formation is usually preceded by hyperfluorescence of photoproducts and weak plasma luminescence through Bremsstrahlung from CB electrons. All three markers can be used as physical real-time indicators for free-electron-mediated chemical changes [407,409].

A mechanistic understanding of such low-density plasma effects can be achieved by numerical simulations of plasma formation that consider the plasma initiation channels constituted by the specific band structure of water and by biomolecules with low ionization energy [36,37,196-198]. A key feature governing the interaction between electrons and biomolecules in the low-density plasma regime is the kinetic energy spectrum of the laser-induced conduction band electrons [149], which depends strongly on irradiance and laser wavelength [198]. An essential prerequisite for its calculation is the consideration of the stepwise gain of kinetic energy of CB electrons by inverse bremsstrahlung absorption, which can be achieved by solving a multi-rate equation [149,198]. Knowledge of the kinetic energy spectrum enables to tune the biomolecular effects by appropriate choice of the laser parameters, if the action spectra for different chemical changes as a function of kinetic electron energy are known [198,203,412,413].

Free-electron-mediated chemical changes with control of the kinetic energy spectrum of CB electrons through appropriate choice of laser parameters make it possible to induce specific modifications of various types of biomolecules. Examples are the induction of different damage types in DNA for studying repair processes in living cells [198,203,414], gentle fragmentation of large protein segments for improving resolution in mass spectroscopy [415,416], and the induction of local refractive index changes in the cornea for the correction of myopia or presbyopia [17,417]. Life-cell experiments with

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tailored electron energy spectra can provide a better understanding of LEE-induced reaction pathways in targeted laser nanosurgery and radiation therapy [198].

Free-electron-mediated molecular disintegration and chemical changes play a role also in the transition zone from optical breakdown involving phase transitions by single pulses to low-density plasma formation with gas bubble formation by multiple pulses. Micro-and nanosurgical applications in the transition zone include corneal dissection by low-energy fs pulses applied at MHz repetition rate [17] and gentle optoporation of cells mediated by molecularly targeted gold NP that are irradiated by series of off-resonance fs pulses [418-420]. Reactive LAL in organic solvents for NP synthesis and tailoring is also a mixture of thermodynamic events, surface chemistry, and free-electron-mediated chemical effects [98,100]. The same applies for plasma formation by electric discharge at liquid surfaces and within liquids [271,421-423].

Altogether, the energy dependence of plasma formation in dielectrics comprises a very large range of laser pulse energies and plasma energy densities, and the laser-induced effects exhibit an enormous variety. The results and the framework for the analysis presented in this paper may help to guide future mechanistic analysis and stimulate the exploration of novel applications reaching from free-electron-mediated molecular modifications to the generation and characterization of extreme states of matter. A better understanding of the complete range of plasma-induced processes from the thermomechanical effects to free-electron-mediated changes opens new pathways for laser surgery, material processing, material research, various sustainable technologies, and can even improve nonlinear microscopy.

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